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17. Disinfectants An experiment was conducted to study the use of 95% ethanol versus 20% bleach as a disinfectant in removing contamination when culturing plant tissues. The experiment was repeated 15 times with each disinfectant, using cuttings of eggplant tissue.[9] The observation reported was the number of uncontaminated eggplant cuttings after a 4-week storage.

Disinfectant	95% Ethanol	20% Bleach
Mean	3.73	4.80
Variance	2.78095	.17143
n	15	15
	Pooled variance 1.47619	

a. Are you willing to assume that the underlying variances are equal?

b. Using the information from part a, is there evidence of a significant difference in the mean numbers of uncontaminated eggplants for the two disinfectants tested?

a) $H_0: \sigma_1^2 = \sigma_2^2$

$H_a: \sigma_1^2 \neq \sigma_2^2$

$$F = \frac{S_1^2}{S_2^2} = \frac{2.78095}{0.17143} = 16.22$$

$$t_{df_1, df_2, \frac{\alpha}{2}} = t_{14, 14, 0.025} = 2.9786$$

$\because 16.22 > 2.9786$

$\therefore$ Reject H_0

b) $H_0: \mu_1 - \mu_2 = 0$

$H_a: \mu_1 - \mu_2 \neq 0$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{3.73 - 4.80}{\sqrt{\frac{2.8095}{15} + \frac{0.17143}{15}}} = \frac{-1.07}{0.4437} = -2.41$$

$$df = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)}{\frac{\left(\frac{S_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{S_2^2}{n_2}\right)^2}{n_2 - 1}} = \frac{(0.1854 + 0.0114)^2}{\frac{0.1854^2}{14} + \frac{0.0114^2}{14}} = \frac{0.03873}{0.002467} = 15.72 \approx 16$$

$$|t| = t_{df, \frac{\alpha}{2}} = t_{16, 0.025} = 2.12$$

$\because |-2.41| = 2.41 > 2.12$

$\therefore$ Reject H_0

20. Dissolved O_2 Content A pollution control inspector suspected that a river community was releasing amounts of semitreated sewage into a river. To check his theory, he drew five randomly selected specimens of river water at a location above the town, and another five below. The dissolved oxygen readings (in parts per million) are as follows:

Above Town	4.8	5.2	5.0	4.9	5.1
Below Town	5.0	4.7	4.9	4.8	4.9

a. Do the data provide sufficient evidence to indicate that the mean oxygen content below the town is less than the mean oxygen content above? Test using $\alpha = .05$.

b. Estimate the difference in the mean dissolved oxygen contents for locations above and below the town. Use a 95% confidence interval.

a) $H_0: \mu_B - \mu_A = 0$

$H_a: \mu_B - \mu_A < 0$

$$S_p^2 = \frac{(n_A - 1)S_A^2 + (N_B - 1)S_B^2}{N_A + N_B - 2}$$

$$= \frac{4(0.025) + 4(0.073)}{5+5-2}$$

$$= 0.019$$

$$S_p = 0.1378$$

$$t = \frac{\bar{X}_B - \bar{X}_A}{S_p\sqrt{\frac{1}{n_B} + \frac{1}{N_A}}}$$

$$= \frac{4.86 - 5}{0.1378\sqrt{\frac{1}{5} + \frac{1}{5}}}$$

$$= -1.606$$

$df = N_A + N_B - 2$

$= 8$

$t_{df,\alpha} = t_{8,0.95}$

$= -1.86$

$\because -1.606 > -1.86$

$\therefore$ Non-reject H_0

$$\bar{X}_A = \frac{4.8 + 5.2 + 5.0 + 4.9 + 5.1}{5}$$

$= 5$

$$S_A^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}$$

$$= \frac{0.1}{4}$$

$= 0.025$

$$\bar{X}_B = \frac{5 + 4.7 + 4.9 + 4.8 + 4.9}{5}$$

$= 4.86$

$$S_B^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}$$

$$= \frac{0.05^2}{4}$$

$= 0.013$

b) $(\bar{X}_A - \bar{X}_B) \pm t_{8,0.025} \cdot S_p\sqrt{\frac{1}{n_A} + \frac{1}{N_B}}$

$= 5 - 4.86 \pm 2.306(0.08718)$

$= 0.14 \pm 0.201$

$= (-0.067, 0.341)$

$\therefore$ The 95% CI is $(-0.067, 0.341)$ ppm

Paired-Difference Tests II *Use the information in Exercise 9 to calculate the observed value of the t statistic for testing the difference between the two population means using paired data. Find a rejection region with $\alpha = .05$ to state your conclusions. Construct a 95% confidence interval for $\mu_1 - \mu_2$. Does this confirm the results of your hypothesis test?*

9.

	Pairs				
Population	**1**	**2**	**3**	**4**	**5**
1	1.3	1.6	1.1	1.4	1.7
2	1.2	1.5	1.1	1.2	1.8

Pair	Pop 1	Pop 2	d
1	1.3	1.2	0.1
2	1.6	1.5	0.1
3	1.1	1.1	0
4	1.4	1.2	0.2
5	1.7	1.8	-0.1

$H_0 : \mu_d = 0$

$H_a : \mu_d \neq 0$

$$\bar{d} = \frac{0.1 + 0.1 + 0.4 + 0.2 + 1 - 0.1}{5}$$

$= 0.06$

$$S_d = \sqrt{\frac{\sum_{i=1}^{n} (d_i - \bar{d})^2}{n-1}}$$

$= \sqrt{\frac{0.052}{4}}$

$= \sqrt{0.013}$

$= 0.114$

$$SE = \frac{S_d}{\sqrt{n}}$$

$= \frac{0.114}{\sqrt{5}}$

$= 0.051$

$$t = \frac{\bar{d} - 0}{SE}$$

$= \frac{0.06}{0.051}$

$= 1.18$

$df = n - 1$

$= 4$

$t_{df, \frac{\alpha}{2}} = t_{4, 0.025}$

$= 2.776$

$\because 1.18 < 2.776$

$\therefore$ Non-reject H_0

$\bar{d} \pm t_{4, 0.025} \cdot SE = 0.06 \pm 2.776 (0.051)$

$= 0.06 \pm 0.1416$

$= (-0.0816, 0.2016)$

DATA SET DS1026

20. Memory Experiments Twenty students participated in an experiment in which they were first asked to recall a list of 25 words. They were then given a second list, and asked to form images of the words as they were read. The number of words recalled for each student is given in the table.

Student	With Imagery	Without Imagery
1	20	5
2	24	9
3	20	5
4	18	9
5	22	6
6	19	11
7	20	8
8	19	11
9	17	7
10	21	9
11	17	8
12	20	16
13	20	10
14	16	12
15	24	7
16	22	9
17	25	21
18	21	14
19	19	12
20	23	13

Does it appear that the average recall score is higher when imagery is used?

$H_0 : \mu_d = 0$

$H_a : \mu_d \neq 0$

$n = 20$

$$\bar{d} = \frac{15+15+15+9+16+8+12+8+10+12+11+4+10+4+17+13+4+7+17+10}{20}$$

$= 0.25$

$$s_d = \sqrt{\frac{\sum_{i=1}^{n} (d_i - \bar{d})^2}{n-1}}$$

$= 4.651$

$$SE = \frac{s_d}{\sqrt{n}}$$

$= \frac{4.651}{\sqrt{20}}$

$= 0.906$

$$t = \frac{\bar{d} - 0}{SE}$$

$= \frac{10.25}{0.906}$

$= 11.31$

$df = n - 1$

$= 19$

$t_{df,\alpha} = t_{19, 0.05}$

$= 1.729$

$\because 11.32 > 1.729$

$\therefore$ Reject H_0

$\bar{d} \pm t_{19, 0.05} \cdot SE = 10.25 \pm 2.093 (0.906)$

$= 10.25 \pm 1.896$

$= (8.354, 12.146)$

DATA SET DS1028

22. Cake Mixes An experiment was conducted to compare the densities (in ounces per cubic inch) of cakes prepared from two different cake mixes. Six cake pans were filled with batter A, and six were filled with batter B. Expecting a variation in oven temperature, the experimenter placed a pan filled with batter A and another with batter B *side by side* at six different locations in the oven. The six paired observations of densities are as follows:

Location	1	2	3	4	5	6
Batter A	.135	.102	.098	.141	.131	.144
Batter B	.129	.120	.112	.152	.135	.163

a. Do the data present sufficient evidence to indicate a difference between the average densities of cakes prepared using the two types of batter?

b. Construct a 95% confidence interval for the difference between the average densities for the two mixes.

Location	A	B	d
1	0.135	0.129	0.006
2	0.102	0.120	−0.018
3	0.098	0.112	−0.014
4	0.141	0.152	−0.011
5	0.131	0.135	−0.004
6	0.144	0.163	−0.019

a) $\bar{d} = \dfrac{0.006 + (-0.018) + (-0.011) + (-0.004) + (-0.019)}{6}$

$= -0.01$

$S_d = \sqrt{\dfrac{\sum_{i=1}^{n} (d_i - \bar{d})^2}{n-1}}$

$= 0.00953$

$H_0: \mu_d = 0$

$H_a: \mu_d \neq 0$

$SE = \dfrac{S_d}{\sqrt{n}}$

$= \dfrac{0.00953}{\sqrt{6}}$

$= 0.00389$

$t = \dfrac{\bar{d} - 0}{SE}$

$= \dfrac{-0.01}{0.00389}$

$= -2.57$

$df = n - 1$

$= 5$

$t_{df, \frac{\alpha}{2}} = t_{5, 0.025}$

$= 2.571$

$\because |t| \approx 2.57 < 2.571$

$\therefore$ Non-reject H_0

b) $\bar{d} \pm t_{5, 0.025} \cdot SE = -0.01 \pm 2.571 (0.00389)$

$= -0.01 \pm 0.01$

$= (-0.02, 0)$

DATA SET DS1031

12. A Precise Quarterback The number of passes completed and the total number of passing yards recorded for the Los Angeles Chargers quarterback, Philip Rivers for each of the 15 regular season games that he played in the fall of 2017[13] were used to calculate the average number of yards per pass for each game.

Week	Completions	Yards per Pass	Week	Completions	Yards per Pass
1	28	13.8	9	17	12.5
2	22	13.2	11	15	12.2
3	20	11.4	12	25	10.7
4	18	17.7	13	21	12.3
5	31	11.1	14	22	15.8
6	27	16.1	15	20	11.8
7	20	12.6	16	31	10.7
8	21	17.2	17	22	8.7

a. Calculate the mean and standard deviation of the number of completions and the number of yards per pass.

b. Find a 95% confidence interval estimate for the variance of the number of completions. Why would you prefer a small variability in the number of completed passes?

c. Find a 95% confidence interval for σ^2, the variance of the yards per pass. Use these results to find a 95% confidence interval for σ, the standard deviation of the yards per pass.

d. Test whether the standard deviation of the yards per pass for this quarterback differs from $\sigma = 4$ with $\alpha = .05$.

a) Completions

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{360}{16} = 22.5$$

$$s = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1} = 4.705$$

Yards per pass

$$\bar{x} = \frac{\sum x_i}{n} = \frac{207.8}{16} = 12.9875$$

$$s = \frac{\sum (x_i - \bar{x})^2}{n-1} = 2.531$$

b) $\alpha = 0.05$, $\frac{\alpha}{2} = 0.025$, $df = n-1 = 15$

$$\text{Completions } 95\% \text{ CI} = \left[\frac{(n-1)s^2}{\chi_{15,0.025}}, \frac{(n-1)s^2}{6.2621}\right]$$

$$= \left[\frac{15(4.705)^2}{27.488}, \frac{15(4.705)^2}{6.2621}\right]$$

$$= (12.0798, 53.0262)$$

∴ I prefer that because it shows that the quarterback's performance is more consistent and predictable.

c) $s^2 = 6.406$

Yards per pass 95% CI

$$= \left[\frac{(n-1)s^2}{\chi^2_{15,0.025}}, \frac{(n-1)s^2}{\chi^2_{15,0.975}}\right]$$

$$= \left[\frac{15(6.406)}{27.488}, \frac{15(6.406)}{6.2621}\right]$$

$$= (3.4957, 15.3447)$$

∴ 95% CI for σ^2: $(3.4957, 15.3447)$

$$\sigma \in (\sqrt{3.4957}, \sqrt{15.3447}) = (1.8756, 3.9216)$$

∴ 95% CI for σ: $(1.8697, 3.9172)$

d) $H_0: \sigma = 4$ $(\sigma^2 = 16)$

$H_a: \sigma \neq 4$

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{15(6.406)}{16} = 6.0056$$

$df = n-1 = 15$

Rejection Region $= [\chi^2_{15,0.975}, \chi^2_{15,0.05}] = (6.262, 27.488)$

∵ $6.0056 < 6.262$

∴ Reject H_0

13. Instrument Precision A precision instrument is guaranteed to read accurately to within 2 units. A sample of four instrument readings on the same object yielded the measurements 353, 351, 351, and 355.

a. Test the null hypothesis that $\sigma = .7$ against the alternative $\sigma > .7$. Use $\alpha = .05$.

b. Find a 90% confidence interval for the population variance.

$$\bar{X} = \frac{353 + 351 + 351 + 355}{4}$$

$$= 3525$$

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$$

$$= \frac{11}{3}$$

$$= 3.667$$

$$s = \sqrt{3.667}$$

$$= 1.915$$

a) $H_0: \sigma = 0.7$

$H_a: \sigma > 0.7$

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$$

$$= \frac{3(3.667)}{(0.7)^2}$$

$$= 22.45$$

$df = n-1$
$= 3$

$\chi^2_{3, 0.05} = 7.815$

$\because 22.45 > 7.815$

$\therefore$ Reject H_0

b) σ^2 90% CI $= \left[\frac{11.001}{\chi^2_{3, 0.05}}, \frac{11.001}{\chi^2_{3, 0.95}} \right]$

$$= \left[\frac{11.001}{7.815}, \frac{11.001}{0.352} \right]$$

$$= [1.41, 31.25]$$

$\therefore$ 90% CI for $\sigma^2 = (1.41, 31.25)$